

Major FYP

Datasets

Static Datasets

- ASL alphabets [\[go\]](#) [\[also this\]](#)

Goal: Comparison of performance of multiclass classification and contrastive learning.

Dynamic Datasets

Goals: control the screen (touchless screen control)

- IPN Hand [\[go\]](#)
- JESTER dataset [\[go\]](#)
- (Not accesible yet) DVS128 dataset [\[go\]](#)

Goals: Generalizability over unseen gestures with minimal samples.

- E. Uboweja, D. Tian, Q. Wang, et al., On-device real-time custom hand gesture recognition, 2023. arXiv: 2309. 10858 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2309.10858>.
 - They trained a word-level fingerspelling model which utilizes a single-hand embedding sub-model to extract discriminative features as feature vectors - the weights are the transferred to a new model for training a custom gesture recognition model with minimal training data.

Goals: Generalizability over unseen gestures, including different body parts (including face, etc) - with minimal samples for retraining.

- HKSL reference (<http://www.cslds.org/v4/>)
 - Sign language investigation: <https://www.sciencedirect.com/science/article/pii/S0024384121001601>
 - Hard negatives & Phrasal separation (CANNOT (ability) vs CANNOT (permission))
 - Adapting to varying speeds of making the signs.
 - Names and context getting.

Goals: Sign language interpretation without the need to pretrain every gloss.

Goals: Extract extra information - like **distance**, **direction**, **object in contact**, etc. from the gesture.

Techniques

- augments: angle (within 20 degrees difference)
- keypoints extraction

Checkpoints

- Before 23 Sept: Multiclass on ASL, Constrastive on ASL.

Objectives